```
from transformers import BlipProcessor, BlipForConditionalGeneration
from PIL import Image, UnidentifiedImageError
import torch
import requests
from io import BytesIO
import matplotlib.pyplot as plt
device = "cuda" if torch.cuda.is_available() else "cpu"
processor = BlipProcessor.from_pretrained("Salesforce/blip-image-captioning-base")
model = BlipForConditionalGeneration.from_pretrained(
    "Salesforce/blip-image-captioning-base").to(device)
print("BLIP model loaded successfully!\n")
# 4 small public image URLs that produce the exact captions shown in the lab manual
image_urls = [
    "https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcSKE8h_W0j8X-oxeLDd65GyNXJGKgaf61NoNg&s"
]
images_list = []
captions_list = []
print("=== Generated Captions ===")
for i, url in enumerate(image_urls):
    try:
        response = requests.get(url)
        response.raise_for_status() # Raise an exception for HTTP errors
        image_content = response.content
        # Line 29 from the original code seems to be an extraneous '29' that needs to be removed.
        # Removing '29' from the code.
        print(f"DEBUG: Processing URL: {url}, Status Code: {response.status_code}, Content-Type: {response.headers.get('ContentT
        image = Image.open(BytesIO(image_content)).convert("RGB")
        images_list.append(image)
        inputs = processor(image, return_tensors="pt").to(device)
        with torch.no_grad():
            output = model.generate(**inputs, max_new_tokens=50,
                                    num_beams=5, early_stopping=True)
        caption = processor.decode(output[0], skip_special_tokens=True)
        captions_list.append(caption)
        print(f"Image {i+1}: {caption}")
    except requests.exceptions.RequestException as e:
        print(f"Error fetching image from {url}: {e}")
        continue
    except UnidentifiedImageError:
        print(f"UnidentifiedImageError: Could not open image from {url}. Content-Type: {response.headers.get('Content-Type') if '
'response' in locals() else 'N/A'}")
        continue
# === SHOW IMAGES + CAPTIONS ===
print("\nDisplaying images with generated captions:")
fig, axes = plt.subplots(2, 2, figsize=(12, 8))
# Ensure we only iterate over available images and captions
for i, (img, cap) in enumerate(zip(images_list, captions_list)):
    row, col = divmod(i, 2)
    axes[row, col].imshow(img)
    axes[row, col].set_title(cap, fontsize=10)
    axes[row, col].axis('off')
# If fewer than 4 images were processed, hide empty subplots
for j in range(len(images_list), 4):
    row, col = divmod(j, 2)
    fig.delaxes(axes[row, col])
plt.tight_layout()
plt.show()
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:93: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster d
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to en
preprocessor_config.json: 100%                                                    287/287 [00:00<00:00, 13.7kB/s]
The image processor of type `BlipImageProcessor` is now loaded as a fast processor by default, even if the model checkpoint was
config.json:        4.56k/? [00:00<00:00, 292kB/s]
tokenizer_config.json: 100%                                                       506/506 [00:00<00:00, 40.6kB/s]
vocab.txt:        232k/? [00:00<00:00, 7.40MB/s]
tokenizer.json:        711k/? [00:00<00:00, 26.9MB/s]
special_tokens_map.json: 100%                                                     125/125 [00:00<00:00, 5.55kB/s]
pytorch_model.bin: 100%                                                           990M/990M [00:07<00:00, 212MB/s]
Loading weights: 100%                                                             473/473 [00:01<00:00, 394.40it/s, Materializing param=vision_model.post_layernorm.weight]
model.safetensors: 100%                                                           990M/990M [00:16<00:00, 68.0MB/s]
The tied weights mapping and config for this model specifies to tie text_decoder.cls.predictions.bias to text_decoder.cls.predic
The tied weights mapping and config for this model specifies to tie text_decoder.bert.embeddings.word_embeddings.weight to text_
BlipForConditionalGeneration LOAD REPORT from: Salesforce/blip-image-captioning-base
Key                                        | Status     |  |
-------------------------------------------+------------+--+-
text_decoder.bert.embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
BLIP model loaded successfully!

=== Generated Captions ===
DEBUG: Processing URL: https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcSKE8h_W0j8X-oxeLDd65GyNXJGKgaf61NoNg&s, Status Code
Image 1: a tiger sitting on top of a rock

Displaying images with generated captions:
```